

Scenario no.1(wired connection/metro exodus/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wired ethernet

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller

Connection bandwidth(download): 5.5-6(Mbps)

Connection bandwidth(upload):0.3-0.4(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: metro exodus

System requirements: intel core i5-4440 , 8GB ram , gtx 670

Testing observations

Input lag: 0.14-0.17 S

User experience: overall great and fun experience, graphically the game looks really good not as good as non cloud but still it's great the loading time into the game is very fast as well it took 10-20S for every game to load in all the testing.

Scenario no.2(wired connection/grip combat racing/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wired ethernet

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller/M&K

Connection bandwidth(download): 7.5-8.6(Mbps)

Connection bandwidth(upload):0.35-0.45(Mbps)

Number of players: 2 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: grip combat racing

System requirements: intel core i3-3220 , 4GB ram , geforce 560

Testing observations

Input lag: 0.1-0.2 S

User experience: fun racing game that works really well in both controls and graphical aspect.

Scenario no.3(wired connection/golazo football league/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wired ethernet

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller

Connection bandwidth(download): 10(Mbps)

Connection bandwidth(upload):0.5-0.54(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.1-0.13 S

User experience: a fun cartoonish football game really that also works really great.

Scenario no.4(wireless connection/metro exodus/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller

Connection bandwidth(download): 7-7.5(Mbps)

Connection bandwidth(upload):0.3-0.35(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: metro exodus

System requirements: intel core i5-4440 , 8gb ram , gtx 670

Testing observations

Input lag: 0.2 S

User experience: almost no difference between the wired and wireless

There is a slightly more input lag that you won't notice it at all.

Scenario no.5(wireless connection/grip combat racing/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller/M&K

Connection bandwidth(download): 8-8.7(Mbps)

Connection bandwidth(upload):0.2-0.3(Mbps)

Number of players: 2 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: grip combat racing

System requirements: intel core i3-3220 , 4GB ram , geforce 560

Testing observations

Input lag: 0.1-0.2 S

User experience: same as the wired connection.

Scenario no.6(wireless connection/golazo football league/computer)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller

Connection bandwidth(download): 7-7.5(Mbps)

Connection bandwidth(upload):0.3-0.35(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.14-0.18 S

User experience: same as the wired connection.

Scenario no.7(wireless connection/metro exodus/mobile)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Client specs: mediatek dimensity 920 , 8gb(ram) , 120hz fhd+ amoled 6.67inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 7.1-9(Mbps)

Connection bandwidth(upload):0.3-0.4(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: metro exodus

System requirements: intel core i5-4440 , 8gb ram , gtx 670

Testing observations

Input lag: 0.1-0.2 S

User experience: the mobile experience feels even better than in the pc since the screen is smaller the picture quality will look more crisp and the touch control is not as good as the M&K but still really playable.

Scenario no.8(wireless connection/grip combat racing/mobile)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Clinet specs: mediatek dimensity 920 , 8gb(ram) , 120hz fhd+ amoled 6.67inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 8.5-9.5(Mbps)

Connection bandwidth(upload):0.36-0.45(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: grip combat racing

System requirements: intel core i3-3220 , 4GB ram , geforce 560

Testing observations

Input lag: 0.1-0.2 S

User experience: better than the pc experience and the touch controlling in the racing games feels as good as the xbox controller.

Scenario no.9(wireless connection/golazo football league)

Testing environment

Test location: onsite

Connection type: private network

Latency: 1ms

Connection media: wireless

Client specs: mediatek dimensity 920 , 8gb(ram) , 120hz fhd+ amoled 6.67inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 11(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core i3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.1-0.2 S

User experience: looks great plays great with both xbox controller and touch control with great picture quality.

Scenario no.10(wireless connection/ metro exodus/computer)

Testing environment

Test location: offsite (Ramadi)

Connection type: private network

Latency: 10ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 6(Mbps)

Connection bandwidth(upload):0.5-0.77(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: metro exodus

System requirements: intel core i5-4440 , 8gb ram , gtx 670

Testing observations

Input lag: 0.1-0.2 S

User experience: looks great plays great almost like the onsite.

Scenario no.11(wireless connection/ grip combat racing/computer)

Testing environment

Test location: offsite (Ramadi)

Connection type: private network

Latency: 10ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) , 144hz fhd+ ips 15inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 9(Mbps)

Connection bandwidth(upload):0.25-0.3(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: grip combat racing

System requirements: intel core i3-3220 , 4GB ram , geforce 560

Testing observations

Input lag: 0.1-0.2 S

User experience: looks great plays great almost like the onsite.

Scenario no.12(wireless connection/golazo football league)

Testing environment

Test location: offsite (Ramadi)

Connection type: private network

Latency: 10ms

Connection media: wireless

Client specs: core i5H(11th) , rtx 3050ti , 16gb(ram) ,
144hz fhd+ ips 15inch

Controller type: xbox series controller/touch

Connection bandwidth(download): 11(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD
4000

Testing observations

Input lag: 0.1-0.2 S

User experience: looks great plays great almost like the onsite.

Scenario no.13(wireless connection/golazo football league)

Testing environment

Test location: offsite (Sina`ah street)

Connection type: private network

Latency (avg): 8ms

Connection media: wireless

Client specs: core i3(2nd), Geforce 410m , 4gb(ram) , 60hz hd ips 15inch

Controller type: xbox series controller/M&K.

Connection bandwidth(download): 11(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.4-0.5 S

User experience: looks good plays great with both xbox controller and keyboard with good picture quality.

Scenario no.14(wireless connection/ grip combat racing/computer)

Testing environment

Test location: offsite (Sina`ah street)

Connection type: private network

Latency (avg): 8ms

Connection media: wireless

Client specs: core i3(2nd), Geforce 410m , 4gb(ram) , 60hz hd ips 15inch

Controller type: xbox series controller/M&K.

Connection bandwidth(download): 9(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.4-0.5 S

User experience: looks good plays great with both xbox controller and keyboard with good picture quality.

Scenario no.15(wireless connection/ grip combat racing/computer)

Testing environment

Test location: offsite (Sina`ah street)

Connection type: private network

Latency (avg): 8ms

Connection media: wireless

Client specs: mediatek dimensity 920, 8gb(ram), 120hz fhd+ amoled 6.67inch

Controller type: xbox series controller/M&K.

Connection bandwidth(download): 9(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps, amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.4-0.5 S

User experience: looks good plays great with both xbox controller and keyboard with good picture quality.

Scenario no.16(wireless connection/golazo football league)

Testing environment

Test location: offsite (Sina`ah street)

Connection type: private network

Latency (avg): 8ms

Connection media: wireless

Client specs: mediatek dimensity 920 , 8gb(ram) , 120hz fhd+ amoled 6.67inch

Controller type: xbox series controller/M&K.

Connection bandwidth(download): 9(Mbps)

Connection bandwidth(upload):0.35-0.38(Mbps)

Number of players: 1 player

Streaming profile: (720p,30fps,amd v520 pro)

Game profile

Game name: golazo football league

System requirements: intel core I3 m380 , 2GB ram , intel HD 4000

Testing observations

Input lag: 0.4-0.5 S

User experience: looks good plays great with both xbox controller and keyboard with good picture quality.

Scenario no.1(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.226

Pinging 85.208.229.226 with 32 bytes of data:
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time=1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62

Ping statistics for 85.208.229.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

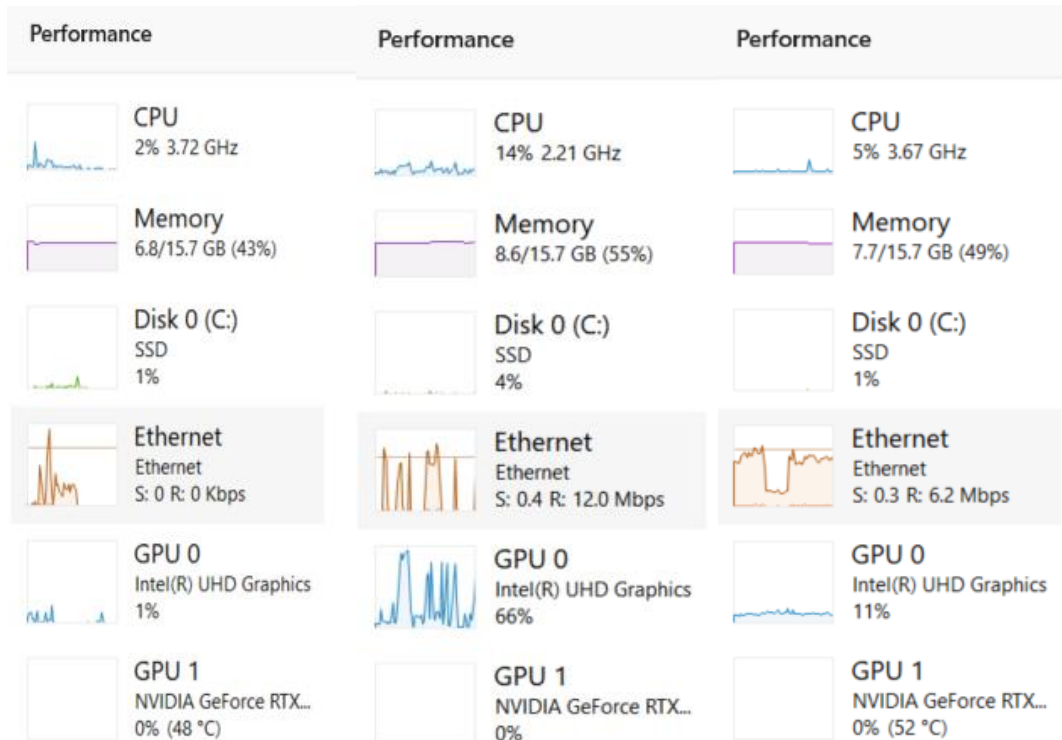
Bandwidth results(pre_launch):

Traffic Total (volume) ⌵	Traffic Total (speed) ⌵	Traffic In (volume) ⌵	Traffic In (speed) ⌵	Traffic Out (volume) ⌵	Traffic Out (speed) ⌵	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⌵	Traffic Total (speed) ⌵	Traffic In (volume) ⌵	Traffic In (speed) ⌵	Traffic Out (volume) ⌵	Traffic Out (speed) ⌵
14 MB	5.76 Mbit/s	13 MB	5.46 Mbit/s	0.71 MB	0.30 Mbit/s
13 MB	5.37 Mbit/s	12 MB	5.06 Mbit/s	0.75 MB	0.32 Mbit/s
17 MB	6.93 Mbit/s	16 MB	6.54 Mbit/s	0.94 MB	0.40 Mbit/s
16 MB	6.84 Mbit/s	16 MB	6.52 Mbit/s	0.77 MB	0.32 Mbit/s
11 MB	4.43 Mbit/s	9.89 MB	4.15 Mbit/s	0.68 MB	0.28 Mbit/s
20 MB	8.30 Mbit/s	19 MB	7.84 Mbit/s	1.10 MB	0.46 Mbit/s

Device usage in metro exodus (heavy_game)



Left: before launching blacknut
 Middle: while launching blakcut
 Right: while launching the game

Scenario no.2(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.226

Pinging 85.208.229.226 with 32 bytes of data:
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time=1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62

Ping statistics for 85.208.229.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

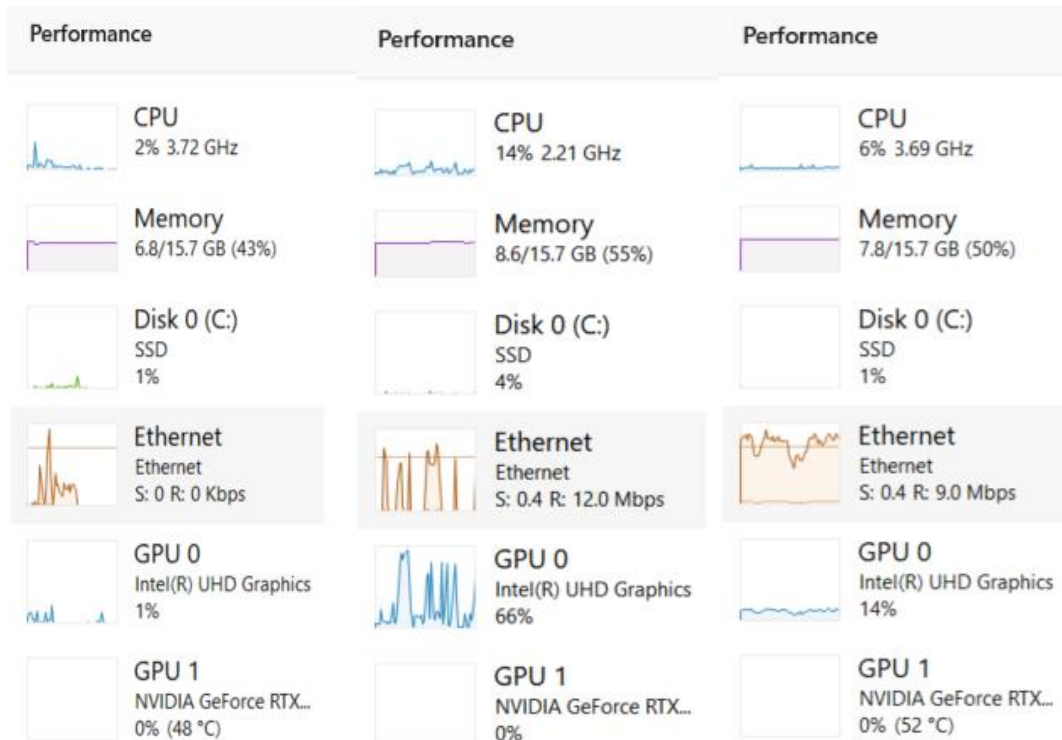
Bandwidth results(pre_launch):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙
22 MB	9.03 Mbit/s	20 MB	8.55 Mbit/s	1.13 MB	0.48 Mbit/s
18 MB	7.58 Mbit/s	17 MB	7.23 Mbit/s	0.83 MB	0.35 Mbit/s
16 MB	6.57 Mbit/s	15 MB	6.19 Mbit/s	0.89 MB	0.37 Mbit/s
20 MB	8.22 Mbit/s	19 MB	7.76 Mbit/s	1.09 MB	0.46 Mbit/s
12 MB	4.91 Mbit/s	11 MB	4.64 Mbit/s	0.64 MB	0.27 Mbit/s
21 MB	8.86 Mbit/s	20 MB	8.38 Mbit/s	1.13 MB	0.47 Mbit/s

Device usage in grip combat racing (moderate_game)



Left: before launching blacknut
 Middle: while launching blakcut
 Right: while launching the game

Scenario no.3(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.226

Pinging 85.208.229.226 with 32 bytes of data:
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62
Reply from 85.208.229.226: bytes=32 time=1ms TTL=62
Reply from 85.208.229.226: bytes=32 time<1ms TTL=62

Ping statistics for 85.208.229.226:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

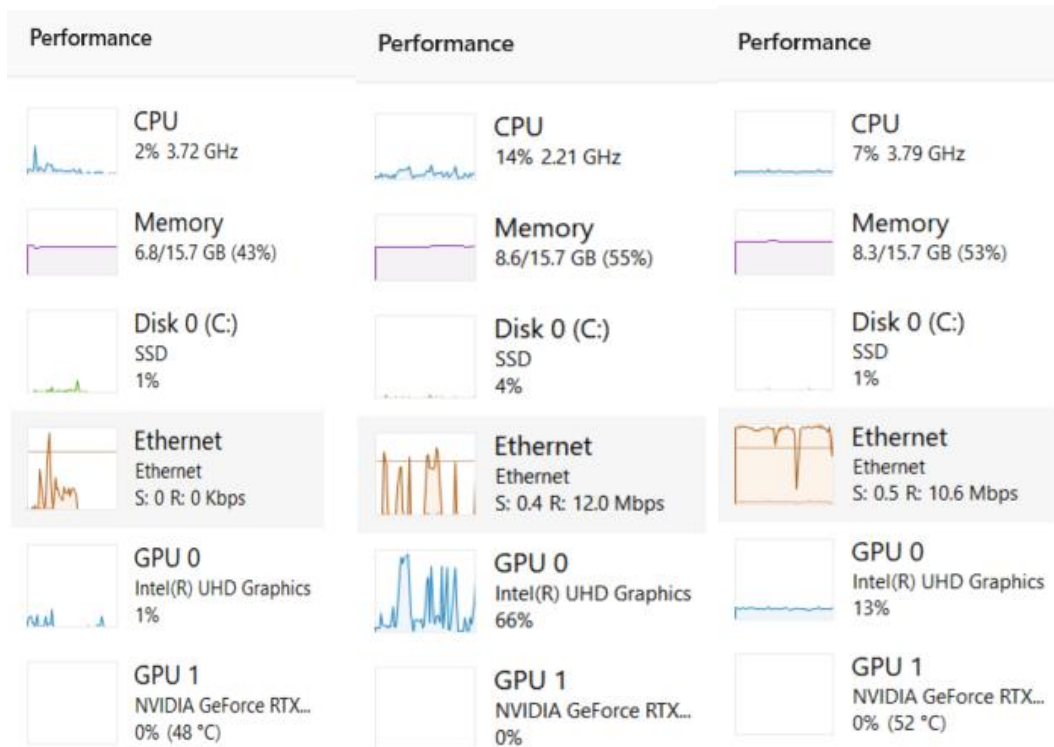
Bandwidth results(pre_launch):

Traffic Total (volume) ▾	Traffic Total (speed) ▾	Traffic In (volume) ▾	Traffic In (speed) ▾	Traffic Out (volume) ▾	Traffic Out (speed) ▾	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ▾	Traffic Total (speed) ▾	Traffic In (volume) ▾	Traffic In (speed) ▾	Traffic Out (volume) ▾	Traffic Out (speed) ▾
26 MB	11 Mbit/s	25 MB	10 Mbit/s	1.35 MB	0.57 Mbit/s
27 MB	11 Mbit/s	25 MB	11 Mbit/s	1.26 MB	0.53 Mbit/s
25 MB	10 Mbit/s	23 MB	9.77 Mbit/s	1.24 MB	0.52 Mbit/s
27 MB	11 Mbit/s	25 MB	11 Mbit/s	1.38 MB	0.58 Mbit/s
26 MB	11 Mbit/s	25 MB	11 Mbit/s	1.25 MB	0.52 Mbit/s
26 MB	11 Mbit/s	25 MB	10 Mbit/s	1.35 MB	0.57 Mbit/s

Device usage in golazo football league (light_game)



Left: before launching blacknut

Middle: while launching blakcut

Right: while launching the game

Scenario no.4(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

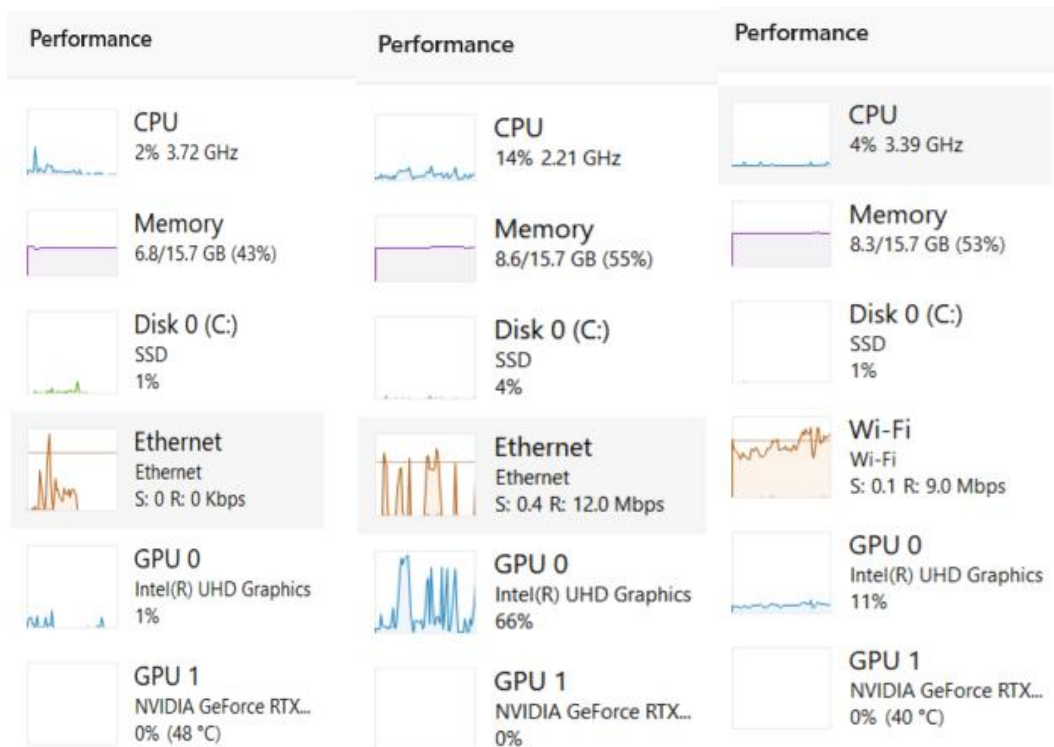
Bandwidth results(pre_launch):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅
18 MB	7.52 Mbit/s	17 MB	7.16 Mbit/s	0.86 MB	0.36 Mbit/s
15 MB	6.30 Mbit/s	15 MB	6.10 Mbit/s	0.47 MB	0.20 Mbit/s
18 MB	7.58 Mbit/s	17 MB	7.27 Mbit/s	0.75 MB	0.32 Mbit/s
19 MB	8.12 Mbit/s	19 MB	7.78 Mbit/s	0.80 MB	0.34 Mbit/s
7.88 MB	3.31 Mbit/s	7.56 MB	3.17 Mbit/s	0.33 MB	0.14 Mbit/s
17 MB	7.09 Mbit/s	16 MB	6.74 Mbit/s	0.84 MB	0.35 Mbit/s

Device usage in metro exodus (heavy_game)



Left: before launching blacknut
 Middle: while launching blakcut
 Right: while launching the game

Scenario no.5(proof)

Latency:


```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

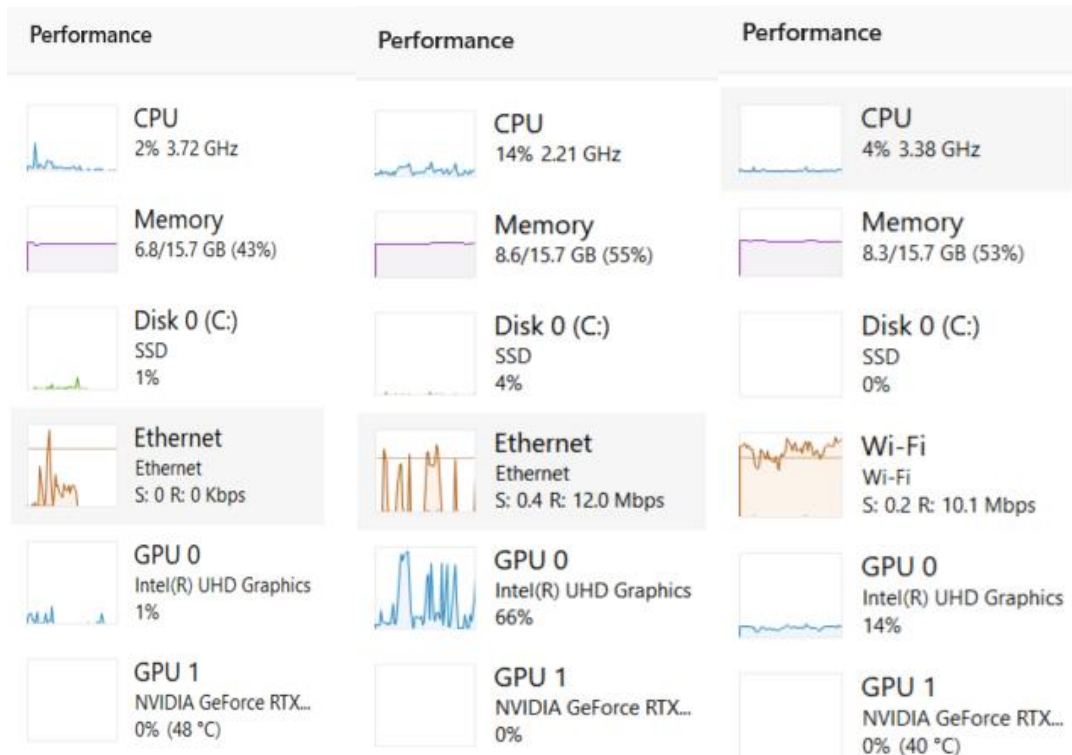
Bandwidth results(pre_launch):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙
21 MB	8.78 Mbit/s	20 MB	8.43 Mbit/s	0.83 MB	0.35 Mbit/s
21 MB	8.93 Mbit/s	21 MB	8.74 Mbit/s	0.45 MB	0.19 Mbit/s
20 MB	8.32 Mbit/s	19 MB	8.12 Mbit/s	0.48 MB	0.20 Mbit/s
20 MB	8.26 Mbit/s	19 MB	7.87 Mbit/s	0.94 MB	0.39 Mbit/s
16 MB	6.61 Mbit/s	15 MB	6.45 Mbit/s	0.39 MB	0.16 Mbit/s
19 MB	7.88 Mbit/s	18 MB	7.67 Mbit/s	0.50 MB	0.21 Mbit/s

Device usage in grip combat racing (moderate_game)



Left: before launching blacknut
 Middle: while launching blakcut
 Right: while launching the game

Scenario no.6(proof)

Latency:

```

Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

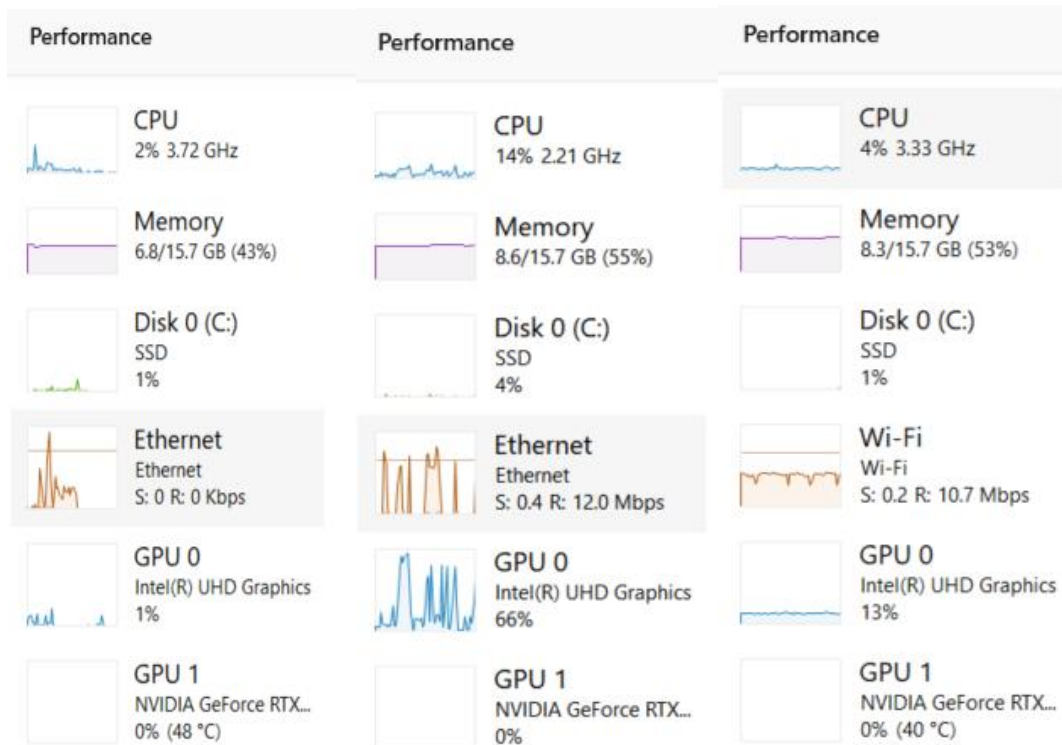
Bandwidth results(pre_launch):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅	Downtime
24 MB	10 Mbit/s	24 MB	9.96 Mbit/s	0.60 MB	0.25 Mbit/s	0 %
25 MB	11 Mbit/s	25 MB	10 Mbit/s	0.54 MB	0.23 Mbit/s	0 %
26 MB	11 Mbit/s	25 MB	11 Mbit/s	0.90 MB	0.38 Mbit/s	0 %
26 MB	11 Mbit/s	25 MB	11 Mbit/s	0.62 MB	0.26 Mbit/s	0 %
26 MB	11 Mbit/s	25 MB	11 Mbit/s	0.83 MB	0.35 Mbit/s	0 %
21 MB	8.76 Mbit/s	20 MB	8.53 Mbit/s	0.54 MB	0.22 Mbit/s	0 %

Device usage in golazo football league (light_game)



Left: before launching blacknut
 Middle: while launching blakcut
 Right: while launching the game

Scenario no.7(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Bandwidth results(pre_launch):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅
18 MB	7.48 Mbit/s	17 MB	7.08 Mbit/s	0.94 MB	0.39 Mbit/s
22 MB	9.15 Mbit/s	21 MB	8.84 Mbit/s	0.73 MB	0.31 Mbit/s
23 MB	9.48 Mbit/s	22 MB	9.13 Mbit/s	0.84 MB	0.35 Mbit/s
20 MB	8.24 Mbit/s	19 MB	7.90 Mbit/s	0.81 MB	0.34 Mbit/s
18 MB	7.49 Mbit/s	17 MB	7.18 Mbit/s	0.74 MB	0.31 Mbit/s
9.54 MB	4 Mbit/s	8.59 MB	3.60 Mbit/s	0.95 MB	0.40 Mbit/s

Scenario no.8(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Bandwidth results(pre_launch):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

Bandwidth results(in_game):

Traffic Total (volume) ⚙	Traffic Total (speed) ⚙	Traffic In (volume) ⚙	Traffic In (speed) ⚙	Traffic Out (volume) ⚙	Traffic Out (speed) ⚙
24 MB	9.93 Mbit/s	23 MB	9.46 Mbit/s	1.12 MB	0.47 Mbit/s
22 MB	9.19 Mbit/s	21 MB	8.76 Mbit/s	1.04 MB	0.43 Mbit/s
22 MB	9.43 Mbit/s	22 MB	9.06 Mbit/s	0.88 MB	0.37 Mbit/s
21 MB	8.90 Mbit/s	20 MB	8.53 Mbit/s	0.86 MB	0.36 Mbit/s
22 MB	9.09 Mbit/s	21 MB	8.72 Mbit/s	0.87 MB	0.37 Mbit/s
22 MB	9.02 Mbit/s	20 MB	8.50 Mbit/s	1.22 MB	0.51 Mbit/s

Scenario no.9(proof)

Latency:

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1265]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hasan>ping 85.208.229.130

Pinging 85.208.229.130 with 32 bytes of data:
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128
Reply from 85.208.229.130: bytes=32 time<1ms TTL=128

Ping statistics for 85.208.229.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

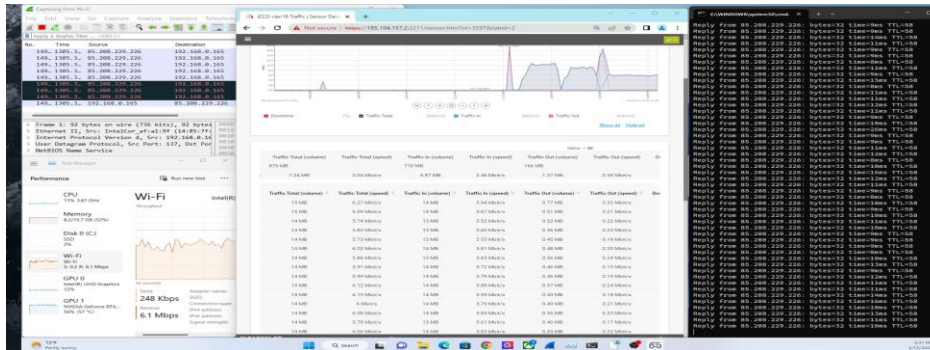
Bandwidth results(pre_launch):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅	Downtime
0.12 MB	0.05 Mbit/s	0.06 MB	0.02 Mbit/s	0.06 MB	0.03 Mbit/s	0 %
0.32 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.25 MB	0.10 Mbit/s	0 %
0.22 MB	0.09 Mbit/s	0.10 MB	0.04 Mbit/s	0.11 MB	0.05 Mbit/s	0 %
0.08 MB	0.04 Mbit/s	0.05 MB	0.02 Mbit/s	0.04 MB	0.02 Mbit/s	0 %
0.31 MB	0.13 Mbit/s	0.07 MB	0.03 Mbit/s	0.24 MB	0.10 Mbit/s	0 %
0.18 MB	0.07 Mbit/s	0.09 MB	0.04 Mbit/s	0.09 MB	0.04 Mbit/s	0 %

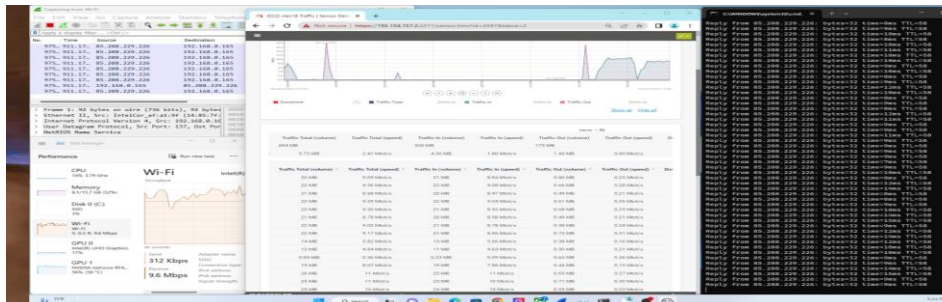
Bandwidth results(in_game):

Traffic Total (volume) ⇅	Traffic Total (speed) ⇅	Traffic In (volume) ⇅	Traffic In (speed) ⇅	Traffic Out (volume) ⇅	Traffic Out (speed) ⇅
26 MB	11 Mbit/s	26 MB	11 Mbit/s	0.83 MB	0.35 Mbit/s
27 MB	11 Mbit/s	26 MB	11 Mbit/s	0.85 MB	0.35 Mbit/s
26 MB	11 Mbit/s	25 MB	11 Mbit/s	0.90 MB	0.38 Mbit/s
26 MB	11 Mbit/s	25 MB	11 Mbit/s	0.85 MB	0.36 Mbit/s
22 MB	9.18 Mbit/s	9.90 MB	4.15 Mbit/s	12 MB	5.03 Mbit/s
0.89 MB	0.37 Mbit/s	0.47 MB	0.20 Mbit/s	0.42 MB	0.18 Mbit/s

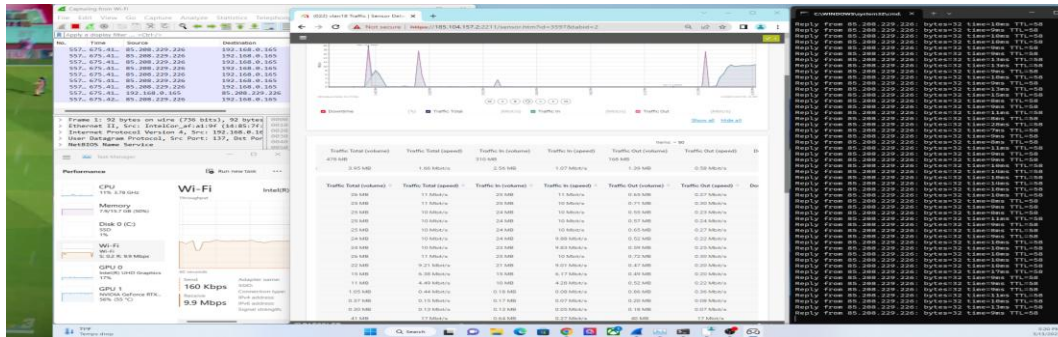
Scenario no.10(proof)



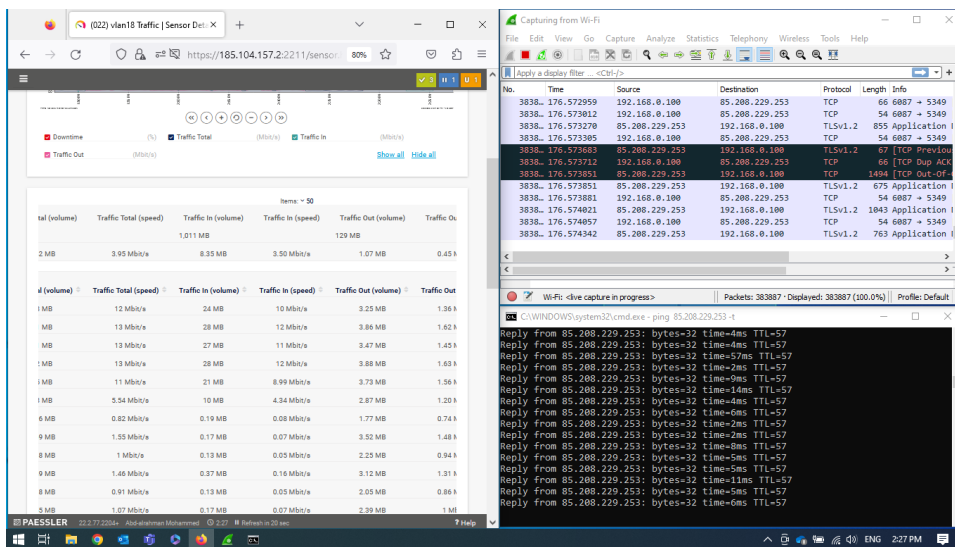
Scenario no.11(proof)



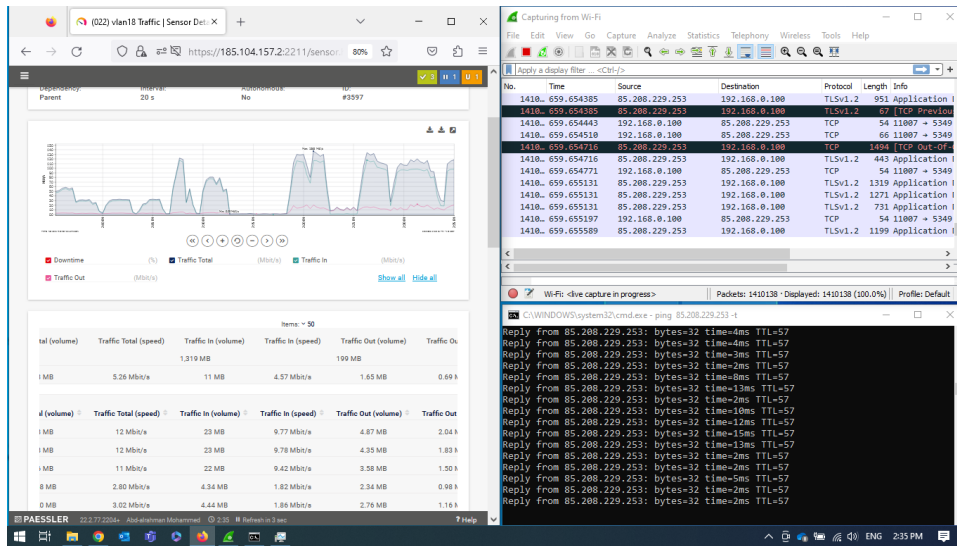
Scenario no.12(proof)



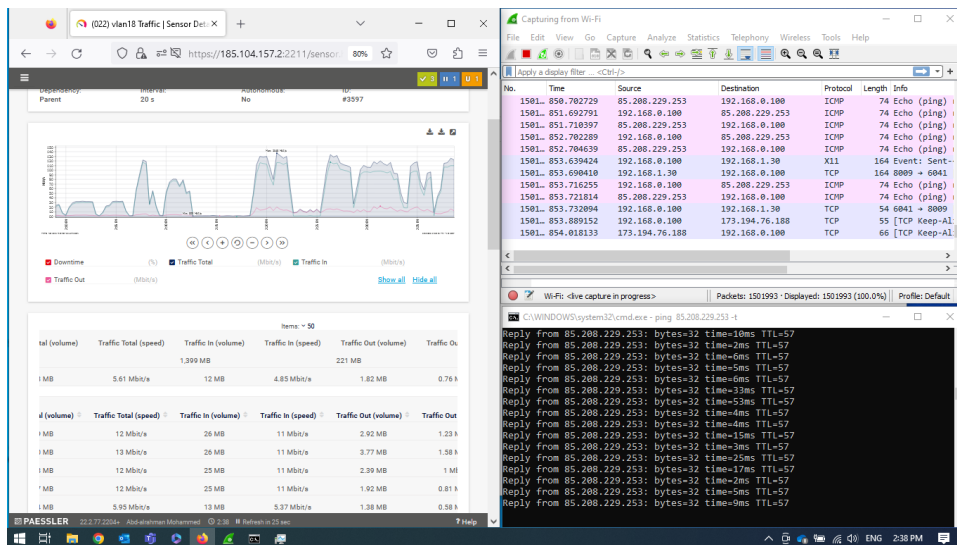
Scenario no.13(proof)



Scenario no.14(proof)



Scenario no.15(proof)



Scenario no.16(proof)

